

BPG-04 · BEST PRACTICE GUIDES

Baselining and baseline change control

What a baseline actually freezes, what the gate has to test,
and why a variance you baseline away is still there

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— Summary

What a project baseline is composed of and what it genuinely freezes, what a change-control gate must test before it approves anything, how re-baselining differs from rolling wave, and why moving a baseline to absorb a variance destroys the only signal the project had. Includes a worked comparison of the correct treatment of an approved scope addition against a re-baseline to forecast, showing what each does to the reported variance and to the funding question underneath it.

— About this document

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Baselining and baseline change control

visible.

In one paragraph. What a project baseline is composed of and what it genuinely freezes, what a change-control gate must test before it approves anything, how re-baselining differs from rolling wave, and why moving a baseline to absorb a variance destroys the only signal the project had. Includes a worked comparison of the correct treatment of an approved scope addition against a re-baseline to forecast, showing what each does to the reported variance and to the funding question underneath it.

Who this is for. Project controls managers and control account managers who maintain a baseline; project managers and sponsors who sit on the change board; anyone being asked to approve a re-baseline and wondering what they are actually approving.

— 1. A baseline is a claim, not a number

A baseline is the project's formal statement that this scope, delivered this way, will take this long and cost this much. Its value comes entirely from being fixed: a number that moves whenever performance disappoints measures nothing, because every comparison against it is a comparison with itself.

That is the whole argument, and it is worth stating before the mechanics, because most baseline failures are not procedural. They happen because somebody found the variance uncomfortable and the baseline was the softest thing in the room.

A useful test for whether a project has a baseline at all: can you produce, in under an hour, the exact version of the scope, schedule and cost plan that the current variance is measured against, together with the list of changes approved into it since sanction? If not, the project has a plan and a forecast but nothing in between.

— 2. What the baseline is composed of

Three components, approved together, each with its own version.

The scope baseline — the work breakdown structure (WBS), the dictionary, and the scope statement it implements. This is the reference for every subsequent question about whether something is a change or was always included. [BPG-02 – The work breakdown structure](#) covers its construction.

The schedule baseline — the network, durations, logic, calendars and constraints as approved, with the schedule basis document that records what the durations assume. A baselined schedule with no basis document is a set of dates whose reasoning has been discarded.

The cost baseline — the budget, time-phased across the schedule at control-account level. This is the planned value curve, and its time-phasing is what makes a cost variance distinguishable from a timing difference.

Two totals sit around these. **Budget at completion** is the sum of the time-phased budgets plus distributed contingency where the organisation's convention places it inside the baseline. Management reserve normally sits outside the performance measurement baseline and is not available to a control account without a formal transfer. Conventions differ legitimately between organisations, so the rule is to state yours in the controls execution plan and apply it consistently rather than to assume; the treatment of both reserves is owned by [BPG-10 – Contingency and management reserve](#).

— 3. What a baseline actually freezes

This is where most disputes originate, because the baseline freezes less than people assume and more than they would like.

It freezes the measurement reference. Every variance reported from now on is measured against this version. That is its primary function.

It freezes the scope commitment. Work not in the scope baseline is, by definition, a change — which is why an ambiguous WBS dictionary entry is a commercial exposure and not just untidiness.

It freezes the time-phasing at control-account level. Not at activity level: activities can and will move within a control account as the schedule is updated, and that movement is progress management, not baseline change. The distinction between "the schedule has been updated" and "the schedule baseline has been changed" is one every planner must be able to state on demand.

It does not freeze the forecast. The forecast is expected to move every period; that is what makes it a forecast. Confusing the two produces the request to "update the baseline to match the forecast", which is §7's subject.

It does not freeze the method. How a work package earns progress can be corrected if the rule was wrong, though the correction must be applied to the whole package and disclosed, because it will move the reported earned value without anything physical happening.

It does not freeze reality. A baseline that has become impossible should be said to be impossible. The correct response is a change request with a defensible recovery plan, not a quiet re-phasing.

— 4. The change-control gate

A change-control gate is not an approval meeting; it is a test. A change request that passes it should have answered six questions in writing, and a board that approves without them is rubber-stamping.

1. **What exactly changes in the scope baseline?** Named WBS elements, with dictionary amendments drafted. "Additional works to the inlet structure" is not an answer.
2. **What is the cost impact, in full?** Direct cost, indirect and time-related cost, escalation on the new scope, and the effect on any allowance. A cost impact stated without its time-related component is usually understated.

3. **What is the schedule impact?** Not "four weeks" but which activities, which logic, and what happens to the critical path and to float elsewhere. A change consuming somebody else's float is still consuming a project asset.
4. **Where does the money come from?** Contingency drawdown, management reserve transfer, client variation, or an unfunded increase in budget at completion. "To be confirmed" means the change is not ready.
5. **What is the impact of not approving it?** The counterfactual is the only way a board can distinguish a genuine requirement from a preference.
6. **What is the effect on reported performance?** Specifically: does approval change any historical variance? The correct answer is almost always no, and if the proposal says yes, §7 applies.

Authority thresholds should be set so that the great majority of changes are approved at the lowest competent level, with a documented escalation. A gate that sends everything to the sponsor is a gate that will be bypassed. [TPL-04 – Baseline change request](#) provides the instrument.

The trend register is the front end of the gate. A trend is a change that has been identified but not yet approved — a design development, a quantity growth, an early warning from a subcontractor. It is logged with an estimated value the moment it is identified, and it appears in the forecast before it appears in the baseline. Projects that log trends only when they become change requests report a forecast that jumps in steps, always upward, always late. [BPG-11 – Change orders and variations](#) covers the commercial instrument; the trend register is the internal one, and it should run ahead of it.

— 5. Rolling wave is not re-baselining

Rolling wave planning holds future scope as **planning packages** — real elements with budget and approximate duration, but no work-package detail — and converts them to detailed work packages as the scope becomes definable.

The conversion moves budget within the baseline. The budget at completion is unchanged, the total time-phased budget is unchanged, and no historical variance is affected. It is a planning act, not a change, and it does not go through the change gate — though it should be recorded, because a conversion that changes the phasing of a planning package's budget will move the planned value curve and therefore the schedule performance index.

Two disciplines keep this honest. The conversion must be **budget-neutral**: if the detailed estimate of the newly defined work exceeds the planning package's budget, that excess is a change and goes through the gate — it does not get absorbed by quietly enlarging the package. And planning packages must have a **conversion deadline**: undefined scope held indefinitely is not rolling wave, it is undistributed budget with no plan, and it hides both scope risk and estimate risk until very late.

— 6. Re-baselining

There are legitimate reasons to re-baseline. A major approved scope change that reshapes the project; a contractual reset; a suspension and restart; an original baseline so far from reality that variance reporting against it has stopped conveying information. In each case the honest test is whether the existing baseline can still support a decision, not whether it is uncomfortable.

A re-baseline done properly has five properties. The old baseline is **archived and remains retrievable**, not overwritten. The **reason is documented** and approved at the level that approved the original. The **cumulative variance to date is stated explicitly** in the approval — the number that is about to disappear from the report is written into the record. The **funding position is resolved**: if the new baseline is larger, the source of the additional funds is named. And **reporting continues against both** for at least one cycle, so readers can see the step.

Where none of those five is present, what is happening is not a re-baseline.

— 7. Keeping the variance visible

The most consequential discipline in this guide is the refusal to let a baseline absorb a performance problem.

The request is always reasonable-sounding. The baseline is out of date; the variance is historical; the team has a credible recovery plan; reporting a large negative variance every month is demoralising and distracts from the work. All of that can be true, and the answer is still no — because the variance is the project's only quantitative record that performance has not matched the plan, and removing it removes the basis for every remaining forecast.

Three consequences follow from baselining a variance away, and §9 demonstrates each.

The forecast loses its evidence. A cost performance index of 0.88 is the empirical basis for expecting the remaining work to cost more than budgeted. Reset it to 1.00 and the forecast has to be justified from optimism instead of from data. [BPG-09 – Estimate at completion](#) depends on this history existing.

The funding gap survives. Moving the baseline changes what the report compares against; it does not change what the project will cost or what has been sanctioned. The difference between the new baseline and the authorised funding is a real gap that now belongs to nobody.

The trend is destroyed. Performance trend — is this getting worse, stable, or recovering? — is the single most useful thing a controls function produces, and it requires an unbroken series. A reset breaks it, and no amount of subsequent reporting reconstructs it.

The alternative is neither difficult nor dramatic: report the variance, report the cause, report the recovery plan and the evidence for it, and let the trend show whether the recovery is happening. A board that sees the same negative variance narrowing over four periods is being told something true. A board that sees it vanish is being told nothing.

— 8. How this goes wrong

The baseline is set before the scope is decomposed. A sanction total is declared the baseline. With no time-phasing, the first variance question — over, or early? — is unanswerable, and the project quietly rebuilds the baseline retrospectively from whatever was spent.

Approved changes are not incorporated. Change requests are approved but never applied to the baseline. The baseline and the authorised scope drift apart, and the reported variance becomes the sum of a real performance problem and an administrative backlog, with no way to separate them.

Historical earned value is restated when scope is added. New scope is added to the baseline and backdated into the earlier periods, so past performance improves without anything having happened. Added scope enters from the approval date forward.

The schedule is re-baselined and the cost baseline is not. The two now describe different plans. Every schedule performance index computed from them is meaningless, and nobody notices for a quarter because the numbers still look like numbers.

Contingency is drawn down without a change record. Money moves from the allowance into a control account to cover an overrun. The control account now shows no variance, the allowance is depleted, and there is no record connecting the two. This is baselining away a variance with extra steps.

"Re-baseline" means "delete the history". The old baseline is overwritten in the tool. Nobody can now state what the project originally committed to, which makes the closeout lessons exercise in [BPG-20](#) impossible to run.

Planning packages never convert. Undefined budget sits in planning packages until late execution, by which time the estimate is worse and the schedule is fixed. The rolling wave discipline exists to prevent exactly this, and it works only with conversion deadlines.

— 9. Worked example

Illustrative figures. Generic currency units. Period: month 8 of a monthly reporting cycle. Values are cumulative to the data date unless stated. Rounding: indices to three decimal places, currency to the nearest 1,000 where a rounded figure is presented. No real project is implied.

9.1 The position at month 8

Measure	Value
Budget at completion (BAC)	6,000,000
Planned value (PV)	2,400,000
Earned value (EV)	2,040,000
Actual cost (AC)	2,310,000

Cost variance (CV), schedule variance (SV), cost performance index (CPI) and schedule performance index (SPI) follow directly:

$$\begin{aligned}
 CV &= EV - AC = 2,040,000 - 2,310,000 = -270,000 \\
 SV &= EV - PV = 2,040,000 - 2,400,000 = -360,000 \\
 CPI &= EV \div AC = 2,040,000 \div 2,310,000 = 0.883 \\
 SPI &= EV \div PV = 2,040,000 \div 2,400,000 = 0.850
 \end{aligned}$$

Forecasting on the assumption that performance to date persists — one method among several, and [BPG-09 – Estimate at completion](#) owns the choice — gives an estimate at completion (EAC) and a variance at completion (VAC):

$$\begin{aligned} \text{EAC} &= \text{BAC} \div \text{CPI} = 6,000,000 \div 0.883116 = 6,794,000 \quad (\text{to the nearest } 1,000) \\ \text{VAC} &= \text{BAC} - \text{EAC} = 6,000,000 - 6,794,000 = -794,000 \end{aligned}$$

The unrounded index $2,040,000 \div 2,310,000 = 0.883116$ is used in the division; using the rounded 0.883 would give 6,795,000, a difference of 1,000 that illustrates why indices are carried unrounded through a calculation and rounded only at presentation.

9.2 An approved scope addition, treated correctly

A change request for additional scope valued at **480,000** is approved at the month-8 board, funded by a client variation. The correct treatment adds the new scope's time-phased budget to the baseline from the approval date forward, and leaves every historical figure alone.

$$\begin{aligned} \text{New BAC} &= 6,000,000 + 480,000 = 6,480,000 \\ \text{EV, AC and the historical CPI are unchanged: CPI remains } 0.883 \\ \text{EAC} &= \text{AC} + (\text{BAC} - \text{EV}) \div \text{CPI} \\ &= 2,310,000 + (6,480,000 - 2,040,000) \div 0.883116 \\ &= 2,310,000 + 4,440,000 \div 0.883116 \\ &= 2,310,000 + 5,028,000 \\ &= 7,338,000 \quad (\text{to the nearest } 1,000) \\ \text{VAC} &= 6,480,000 - 7,338,000 = -858,000 \end{aligned}$$

The forecast overrun has grown from 794,000 to 858,000 — because the added scope is assumed to be delivered at the same efficiency as the work so far, which is an assumption the board should be asked to accept or reject explicitly. The performance signal is intact: CPI is still 0.883, the trend is continuous, and the board can see both that scope was added and that the project is running over.

9.3 The same month, re-baselined to forecast

Now suppose the request is instead to re-baseline the account to the current forecast, resetting earned value to actual cost at the data date so that the account "starts clean".

Measure	Correct treatment (§9.2)	Re-baselined to forecast
BAC	6,480,000	7,338,000
CPI to date	0.883	1.000
CV to date	-270,000	0
VAC	-858,000	0
Forecast cost of the project	7,338,000	7,338,000
Authorised funding	6,480,000	6,480,000
Unresolved funding gap	858,000, reported	858,000, not reported

Both columns describe the same project, costing the same money. Nothing physical differs. What differs is that in the right-hand column the 858,000 gap between what the project will cost and what has been authorised no longer appears anywhere in the performance report — and the 0.883 that was the evidence for expecting it has been replaced by a 1.000 that is evidence of nothing.

Check: $7,338,000 - 6,480,000 = 858,000$, which equals the VAC in the correct treatment. The gap is the same number under both treatments, which is the point.

9.4 Rolling wave, for contrast

The same project holds a planning package of **1,200,000** for later scope. In month 9 that scope becomes definable and is decomposed into five work packages:

$$210,000 + 260,000 + 190,000 + 300,000 + 240,000 = 1,200,000$$

Budget at completion is unchanged at 6,480,000. No historical variance moves. No change request is required, because no scope and no budget has changed — only the level of detail at which it is planned. Had the detailed estimate come to 1,340,000, the excess of $1,340,000 - 1,200,000 = 140,000$ would be a change, and would go through the gate in §4 with its funding source named.

— 10. Checklist

Take this into the change board, or into the meeting where a re-baseline is being proposed.

Before approving any change

- [] Which named WBS elements change, and are the dictionary amendments drafted?
- [] Does the cost impact include time-related cost, escalation on the new scope, and any allowance effect?
- [] Which activities and which logic change, and what happens to the critical path and to others' float?
- [] Where does the money come from — contingency, reserve, client variation, or an unfunded increase?
- [] What happens if this is not approved?
- [] Does approval alter any historical variance? (If yes, ask why before anything else.)

Baseline hygiene

- [] Can the current baseline version and its change history be produced in under an hour?
- [] Have all approved changes actually been incorporated, and how many are outstanding?
- [] Do the schedule baseline and the cost baseline describe the same plan?
- [] Is there a schedule basis document, and was it updated at the last change?
- [] Is management reserve outside the performance measurement baseline, and is the convention written down?

Rolling wave

- [] Does every planning package have a conversion deadline?
- [] Have conversions been budget-neutral, and where they were not, did the excess go through the gate?
- [] What proportion of the budget at completion is still undefined?

If a re-baseline is proposed

- [] Is the old baseline archived and retrievable, or is it about to be overwritten?

- [] Is the cumulative variance being written off stated as a number in the approval?
 - [] Is the funding gap between the new baseline and the authorised amount identified and owned?
 - [] Will both baselines be reported for at least one cycle?
 - [] If the answer to any of the above is no: what is being proposed is not a re-baseline.
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— Related

- [BPG-02 – The work breakdown structure](#) — the scope baseline this depends on, and why dictionary exclusions are a change-control asset.
- [BPG-06 – Progress measurement and rules of credit](#) — the earning rules that make the baseline measurable.
- [BPG-08 – Earned value in practice](#) — how the variances in §9 are computed and read.
- [BPG-10 – Contingency and management reserve](#) — the reserve conventions referenced in §2, and drawdown discipline.
- [BPG-11 – Change orders and variations](#) — the commercial instrument that sits alongside the internal gate.
- [TPL-04 – Baseline change request](#) — the instrument implementing the six-question gate in §4.

— Sources and standards

Drawn from the Institute's Body of Knowledge: Domain 3 (Budgeting and Forecasting) for the time-phased cost baseline and reserves, Domain 5 (Cost Management and Cost Control) for change control and cost impact assessment, Domain 8 (Project Management Lifecycle) for the scope baseline and integrated change control, and Domain 10 (Project Scheduling) for the schedule baseline and its basis document.

The six-question gate in §4 and the five properties of a legitimate re-baseline in §6 are PCI recommended practice. Conventions on whether contingency sits inside or outside the performance measurement baseline differ legitimately between organisations and between published bodies of practice; this guide names the difference rather than asserting one convention as universal. No external standard is reproduced.

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